

EXPERIMENT NO : 09

Experiment to understand Matplotlib library use cases in Data Science through visualization

PROGRAM:-

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

%matplotlib inline

data = pd.read_csv('/content/Iris_Dataset.csv')

data
```

OUTPUT:-

	Id	SepallLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	variety
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa
...	...	...	...	...	...	...
145	146	6.7	3.0	5.2	2.3	Iris-virginica
146	147	6.3	2.5	5.0	1.9	Iris-virginica
147	148	6.5	3.0	5.2	2.0	Iris-virginica
148	149	6.2	3.4	5.4	2.3	Iris-virginica
149	150	5.9	3.0	5.1	1.8	Iris-virginica

150 rows x 6 columns

data.info()

OUTPUT:-

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 6 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Id               150 non-null   int64
1   SepallLengthCm   150 non-null   float64
2   SepalWidthCm     150 non-null   float64
3   PetalLengthCm    150 non-null   float64
4   PetalWidthCm     150 non-null   float64
5   variety          150 non-null   object
dtypes: float64(4), int64(1), object(1)
memory usage: 7.2+ KB
```

```
data.describe()
```

OUTPUT:-

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
count	150.000000	150.000000	150.000000	150.000000	150.000000
mean	75.500000	5.843333	3.057333	3.758000	1.199333
std	43.445368	0.828066	0.435866	1.765298	0.762238
min	1.000000	4.300000	2.000000	1.000000	0.100000
25%	38.250000	5.100000	2.800000	1.600000	0.300000
50%	75.500000	5.800000	3.000000	4.350000	1.300000
75%	112.750000	6.400000	3.300000	5.100000	1.800000
max	150.000000	7.900000	4.400000	6.900000	2.500000

```
data.value_counts('variety')
```

OUTPUT:-

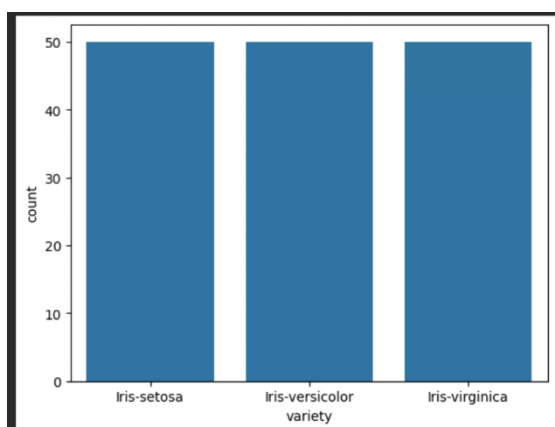
	count
variety	
Iris-setosa	50
Iris-versicolor	50
Iris-virginica	50

dtype: int64

```
sns.countplot(x='variety', data=data)
```

```
plt.show()
```

OUTPUT:-



```
dummies = pd.get_dummies(data.variety)
```

```
dummies
```

OUTPUT:-

	Iris-setosa	Iris-versicolor	Iris-virginica
0	True	False	False
1	True	False	False
2	True	False	False
3	True	False	False
4	True	False	False
...	...	...	...
145	False	False	True
146	False	False	True
147	False	False	True
148	False	False	True
149	False	False	True

150 rows x 3 columns

```
FinalDataset = pd.concat(  
    [pd.get_dummies(data.variety), data.iloc[:, [0,1,2,3]]],  
    axis=1)
```

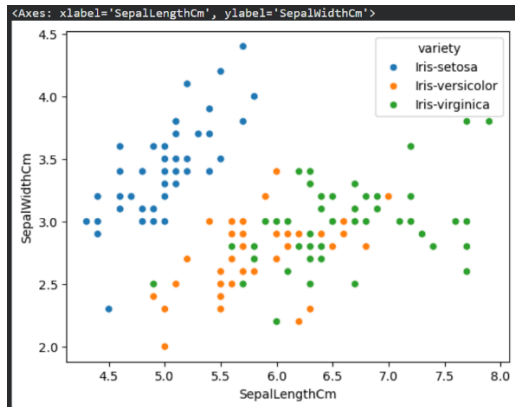
```
FinalDataset.head()
```

OUTPUT:-

	Iris-setosa	Iris-versicolor	Iris-virginica	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm
0	True	False	False	1	5.1	3.5	1.4
1	True	False	False	2	4.9	3.0	1.4
2	True	False	False	3	4.7	3.2	1.3
3	True	False	False	4	4.6	3.1	1.5
4	True	False	False	5	5.0	3.6	1.4

```
sns.scatterplot(x='SepalLengthCm',y='SepalWidthCm',hue='variety',data=ata,)
```

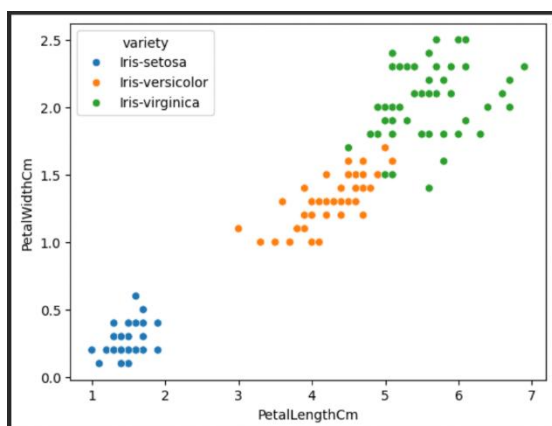
OUTPUT:-



```
sns.scatterplot(  
    x='PetalLengthCm',  
    y='PetalWidthCm',  
    hue='variety',  
    data=data)
```

```
plt.show()
```

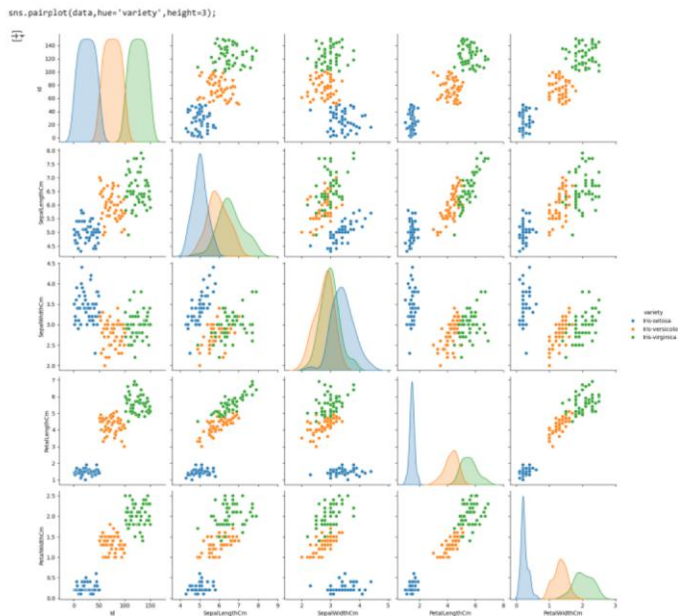
OUTPUT:-



```
sns.pairplot(data, hue='variety', height=3)
```

```
plt.show()
```

OUTPUT:-

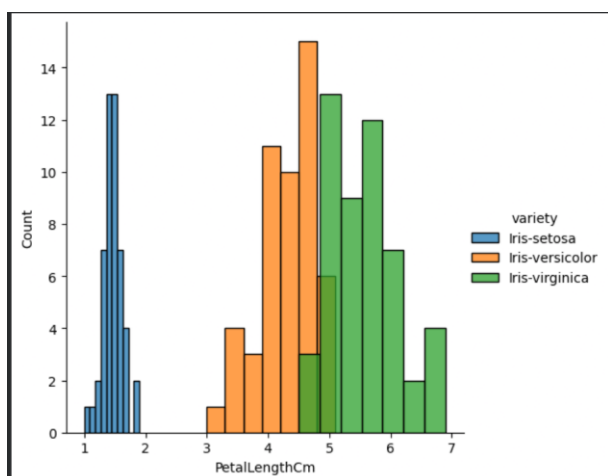


```
sns.FacetGrid(data, hue='variety', height=5).map(
```

```
sns.histplot, 'PetalLengthCm').add_legend()
```

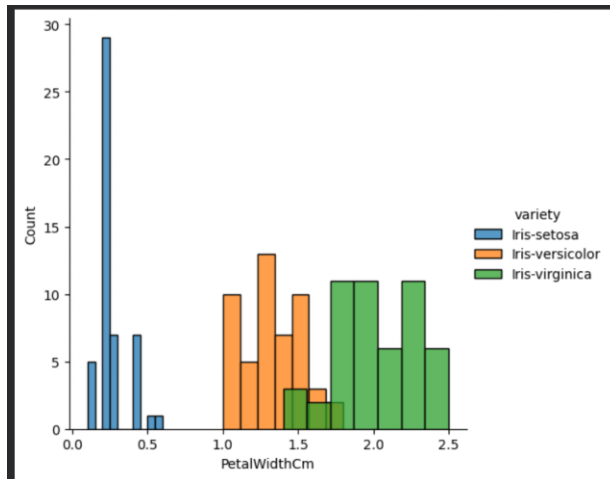
```
plt.show()
```

OUTPUT:-



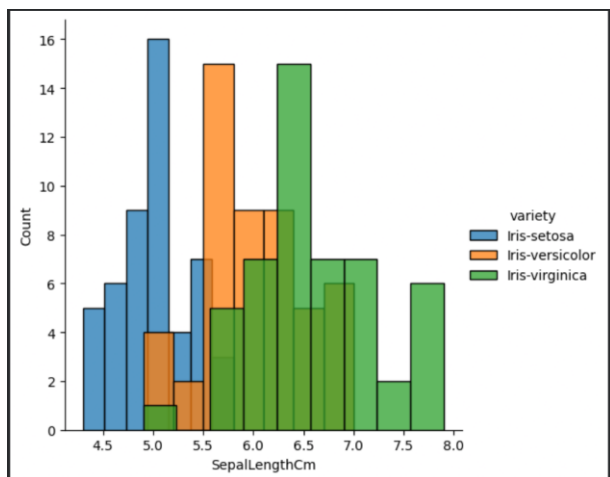
```
sns.FacetGrid(data, hue='variety', height=5).map(  
sns.histplot, 'PetalWidthCm').add_legend()  
  
plt.show()
```

OUTPUT:-



```
sns.FacetGrid(data, hue='variety', height=5).map(  
sns.histplot, 'SepalLengthCm').add_legend()  
  
plt.show()
```

OUTPUT:-



```
sns.FacetGrid(data, hue='variety', height=5).map(  
sns.histplot, 'SepalWidthCm').add_legend()  
plt.show()
```

OUTPUT:-

